



Data Preprocessing in Machine Learning: Cleaning, Scaling, and Leakage

Preprocessing · Machine Learning · R Language



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What is Data Preprocessing?

Before any analysis, data must be prepared. Preprocessing transforms raw data into reliable information — cleaning inconsistencies, integrating distinct sources, and structuring attributes so that mining and machine learning algorithms can operate with maximum effectiveness.

Data Cleaning

Handles missing values, smooths noise, removes outliers, and resolves inconsistencies to ensure data quality.

Data Reduction

Reduces dimensionality and numerosity of data, optimizing processing without losing relevant information.

Data Integration

Consolidates multiple sources — databases, cubes, or files — into a cohesive base ready for analysis.

Data Transformation

Applies normalization, discretization, and concept hierarchies to adapt data to analysis algorithms.

📖 **References:** Han, Pei & Tong (2022). Data Mining: Concepts and Techniques, 4th ed. • García, Luengo & Herrera (2014). Data Preprocessing in Data Mining.

Data Quality: Why Preprocess?

To preprocess data effectively, one must first understand what makes it problematic. Data quality is multidimensional — failures in any of these dimensions compromise analyses and lead to misguided decisions. Understanding them is the starting point for applying the correct preparation techniques.

Accuracy

Do the data correctly reflect reality? Inaccurate values distort any model or analysis.

Consistency

Are the data uniform across sources and records? Internal contradictions invalidate comparisons and aggregations.

Believability

Is the source trustworthy and the collection process robust? The credibility of the data underpins the validity of conclusions.

Completeness

Are all required attributes and records present? Missing values must be addressed before analysis.

Timeliness

Is the information up to date? Outdated data may not represent the actual state of the system.

Interpretability

Are the data understandable? Adequate documentation and metadata facilitate the correct use of information.

 **References:** Han, Pei & Tong (2022). Data Mining: Concepts and Techniques, 4th ed. • García, Luengo & Herrera (2014). Data Preprocessing in Data Mining.

Transformations

Once the dimensions that define data quality are understood, the next step is to act on them. The `daltoolbox` package organizes this work into eight steps that form a cohesive pipeline: starting from sampling and cleaning — which prepare the foundation — through normalizations, encodings, and transformations that make data ready for modeling, up to dimensionality reduction. The balancing and attribute selection steps depend on the concept of class and therefore appear after the initial variable transformations.

Sampling

Train/test split (random and stratified)

Normalization

Normalization of numerical values (min-max and z-score)

Smoothing

Discretization of continuous variables (interval, frequency, and clustering)

Attribute Selection

Identification of relevant predictors (info gain, relief, stepwise, lasso, FSS)

Cleaning

Removal of NAs and outlier detection (boxplot and Gaussian)

Encoding

Transformation of categorical variables (`categ_mapping`)

Balancing

Correction of imbalanced classes (oversampling and undersampling)

Dimensionality Reduction

Compression of the feature space (PCA and curvature heuristics)



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